

Adaptive Fallback as a Witness Stream: The Path-Selection Tax and Honest Tiered Receipts

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Abstract

Tiered deployments are a pragmatic response to the Boss Fight regime: use a cryptographic or delegated *primary* path for most lookups and treat raw DHT lookups as a budgeted *fallback*. A subtle failure mode is that *fallback selection* (when, how often, and for which keys/users a system falls back) is itself a witness stream: if the fallback probability depends on the hidden destination or profile, then the indicator “used fallback” is an identification channel even when each path is individually well-equalized. We formalize this as a simple max-divergence “path-selection tax” and record the minimal receipt fields needed to make tiered designs honest and comparable.

Scope. This note isolates one interface in tiered deployments (Synthesis 10): *path selection*. It is deliberately narrow: we do not propose a concrete caching or routing policy, only a conservative accounting statement that operators can publish.

Imports. We import (i) the tiered primary/fallback pattern and its motivation from Synthesis 10,[3] (ii) Boss Fight budgeting for the fallback path (Boss Fight A/B),[6, 7] and (iii) receipt line-item conventions from the receipt schema note (Synthesis 12).[2]

Exports. A one-line bound (Lemma 1) and a receipt-facing checklist for publishing fallback behavior: a bound on how secret-dependent the fallback probability can be, and how to combine that bound with per-path budgets.

Artifact policy. Source-only; any deployment measurements used to estimate fallback rates are artifacts available on request.[1]

1 Model: two paths and a public indicator

We model a tiered lookup as follows. A hidden secret $C \in \mathcal{C}$ indexes the destination/profile to be protected. The system selects a path $P \in \{\text{primary}, \text{fallback}\}$ and then emits a transcript Z according to the chosen path. The observer sees $Y = (P, Z)$.

Definition 1 (Per-path stability (bits)). *For $p \in \{\text{primary}, \text{fallback}\}$, let $\mathcal{L}(Z \mid C, p)$ be the transcript distribution conditioned on using path p . We say path p is (b_p, δ_p) -stable for the secret pair (c_0, c_1) if for all events A ,*

$$\mathbb{P}[Z \in A \mid C = c_1, p] \leq 2^{b_p} \mathbb{P}[Z \in A \mid C = c_0, p] + \delta_p.$$

The parameter b_p is measured in bits and is directly comparable to per-path MaxL line items in receipts.

Let $\pi_c := \mathbb{P}[P = \text{fallback} \mid C = c]$ be the (possibly secret-dependent) fallback probability.

2 A tax for secret-dependent fallback

Lemma 1 (Path-selection tax (bits)). *Fix two secrets c_0, c_1 and let $\pi_c := \mathbb{P}[P = \text{fallback} \mid C = c]$. Assume per-path stability for both paths with parameters $(b_{\text{primary}}, \delta_{\text{primary}})$ and $(b_{\text{fallback}}, \delta_{\text{fallback}})$. Then the combined channel $Y = (P, Z)$ satisfies, for all events E ,*

$$\mathbb{P}[Y \in E \mid C = c_1] \leq 2^{b^*} \mathbb{P}[Y \in E \mid C = c_0] + \delta^*,$$

where

$$b^* = \max \left\{ b_{\text{primary}} + \log_2 \frac{1 - \pi_{c_1}}{1 - \pi_{c_0}}, b_{\text{fallback}} + \log_2 \frac{\pi_{c_1}}{\pi_{c_0}} \right\}, \quad \delta^* = \delta_{\text{primary}} + \delta_{\text{fallback}}.$$

Proof. Decompose the event E by path. Let $E_p := \{z : (p, z) \in E\}$. Then

$$\mathbb{P}[Y \in E \mid c] = (1 - \pi_c) \mathbb{P}[Z \in E_{\text{primary}} \mid c, \text{primary}] + \pi_c \mathbb{P}[Z \in E_{\text{fallback}} \mid c, \text{fallback}].$$

Apply per-path stability to each term and factor the path probabilities:

$$(1 - \pi_{c_1}) \mathbb{P}[Z \in E_{\text{primary}} \mid c_1, \text{primary}] \leq 2^{b_{\text{primary}}} \frac{1 - \pi_{c_1}}{1 - \pi_{c_0}} (1 - \pi_{c_0}) \mathbb{P}[Z \in E_{\text{primary}} \mid c_0, \text{primary}] + (1 - \pi_{c_1}) \delta_{\text{primary}},$$

and similarly for the fallback term:

$$\pi_{c_1} \mathbb{P}[Z \in E_{\text{fallback}} \mid c_1, \text{fallback}] \leq 2^{b_{\text{fallback}}} \frac{\pi_{c_1}}{\pi_{c_0}} \pi_{c_0} \mathbb{P}[Z \in E_{\text{fallback}} \mid c_0, \text{fallback}] + \pi_{c_1} \delta_{\text{fallback}}.$$

Upper-bound $(1 - \pi_{c_1}) \delta_{\text{primary}} \leq \delta_{\text{primary}}$ and $\pi_{c_1} \delta_{\text{fallback}} \leq \delta_{\text{fallback}}$. Finally, note that

$$2^{b_{\text{primary}}} \frac{1 - \pi_{c_1}}{1 - \pi_{c_0}} = 2^{b_{\text{primary}} + \log_2 \frac{1 - \pi_{c_1}}{1 - \pi_{c_0}}}, \quad 2^{b_{\text{fallback}}} \frac{\pi_{c_1}}{\pi_{c_0}} = 2^{b_{\text{fallback}} + \log_2 \frac{\pi_{c_1}}{\pi_{c_0}}},$$

and combine the two scaled nonnegative terms using $\max(a, b)(x + y) \geq ax + by$ with $x, y \geq 0$. $\square$

Remark 1 (Interpretation). *Lemma 1 says: even if each path is well-budgeted, secret-dependent fallback selection adds an unavoidable log-ratio term. If $\pi_{c_1} \gg \pi_{c_0}$ for some pair, then the single bit “used fallback” is already a strong witness. In tiered anonymity designs, controlling this tax is as important as budgeting the fallback transcript itself.*

3 Receipt fields for honest tiered deployments

Synthesis 12 defines a minimal receipt schema for publishing anonymity claims. For tiered systems, the path-selection tax forces one extra disclosure: a bound on how secret-dependent fallback selection can be.

A minimal, audit-friendly approach is to publish:

- a **fallback policy class** (a small machine-checkable description of what triggers fallback: cache miss, rate-limit, indexer unavailability, etc.),
- an **estimated fallback rate** $\hat{\pi}$ under declared workloads, and
- a conservative **secret-dependence bound** of the form

$$\frac{\pi_{c_1}}{\pi_{c_0}} \leq R_{\pi} \quad \text{and} \quad \frac{1 - \pi_{c_1}}{1 - \pi_{c_0}} \leq R_{\bar{\pi}} \quad \text{for all declared secret pairs.}$$

Given per-path budgets (b_p, δ_p) (primary) and (b_f, δ_f) (fallback), an auditor can then compute the combined worst-case budget by substituting $R_{\pi}, R_{\bar{\pi}}$ into Lemma 1:

$$b^* \leq \max\{b_p + \log_2 R_{\bar{\pi}}, b_f + \log_2 R_{\pi}\}, \quad \delta^* = \delta_p + \delta_f.$$

Operational note. Rate limiting interacts with this tax in two ways: it can reduce the Boss Fight repetition window Q for the fallback path (good), but it can also make fallback selection stateful and secret-correlated (bad) if the rate-limit state depends on content popularity, user cohorts, or cache occupancy. When in doubt, treat fallback selection as a first-class witness stream and publish a conservative R_π bound.

4 Related pointers

For observation probability attenuation (relays, stub mixes) and how it changes Boss Fight budgets, see Synthesis 11; for combining multiple witness tiers (contact-set vs timing vs congestion), see Synthesis 15.[4, 5] For a deployment pattern that uses these receipts to make “privacy debt” legible at the primary/fallback boundary, see Synthesis 10.[3]

References

- [1] Anonymous. *Artifact policy (source-only archive)*. Synthesis note, 2026.
- [2] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
- [3] Anonymous. *Tiered deployment pattern: primary cryptographic path with budgeted fallback*. Synthesis note, 2026.
- [4] Anonymous. Observation Attenuation for Repeated-Use Privacy Budgets: Mixture/Subsampling Bounds and a Stub-Mix Interface. Synthesis 11, The-One-True-Archive (2026).
- [5] Anonymous. Tiered Observation Vectors for Anonymous-DHT Receipts: Contact-Set, Timing, and Congestion Tiers with Conservative MaxL Composition. Synthesis 15, The-One-True-Archive (2026).
- [6] Anonymous. Boss Fight Budgets: Maximal Leakage as a Repeated-Use Anonymity Contract. Boss Fight series Paper A, 2026. (This archive.)
- [7] Anonymous. AnonDHT in the Boss Fight Regime: Compiling a MaxL Budget into Lookup Knobs. Boss Fight series Paper B, 2026. (This archive.)